

LangGraph Agents and Supervision — Part 1

Chapter overview: The supervisor pattern routes each user query to a specialized sub-agent using a Pydantic intent classifier, then aggregates partial answers into a single grounded response with LangGraph Send parallelism.

This section expands on the chapter topic with concrete implementation detail, operational guidance, and worked examples drawn from production deployments. Readers should pay attention to configuration defaults, failure modes, and the observability hooks that make the system debuggable under load. Each subsection closes with a checklist that teams can adopt directly in code review. This section expands on the chapter topic with concrete implementation detail, operational guidance, and worked examples drawn from production deployments. Readers should pay attention to configuration defaults, failure modes, and the observability hooks that make the system debuggable under load. Each subsection closes with a checklist that teams can adopt directly in code review. This section expands on the chapter topic with concrete implementation detail, operational guidance, and worked examples drawn from production deployments. Readers should pay attention to configuration defaults, failure modes, and the observability hooks that make the system debuggable under load. Each subsection closes with a checklist that teams can adopt directly in code review.

Key practice: Sub-agent state schemas stay isolated per domain so the RAG, GitHub, and Google graphs can evolve independently and be wrapped by the supervisor without modification to their nodes or edges.

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Review checklist: Checkpointers keyed by deterministic thread identifiers give every run replayable state, which is required for interrupt based human in the loop confirmations on mutating operations.

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LangGraph Agents and Supervision — Part 2

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Vector Stores and Embeddings — Part 1

Chapter overview: Pinecone serverless indexes host high dimensional embeddings with cosine similarity, while Chroma HTTP serves teams that prefer a self hosted collection behind a simple REST deployment on their own cluster.

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Key practice: Qdrant Cloud collections store payload metadata beside each point so that source, page, and chunk filters can narrow retrieval before ranking runs.

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Review checklist: HuggingFace sentence transformers run locally on CPU for zero key demos, and Gemini text embeddings remain an opt in upgrade for hosted quality.

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RAG Retrieval and MMR Ranking — Part 1

Chapter overview: Maximum marginal relevance balances query similarity against diversity so that near duplicate chunks do not crowd out complementary evidence from other pages of the same document collection.

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Key practice: The `fetch_k` candidate pool feeds the MMR selector, and `lambda_mult` tunes the trade off: values near one behave like pure similarity search while values near zero force maximally diverse coverage of the corpus.

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Review checklist: Every retrieved chunk carries source, page, chunk index, and document id metadata, and the faithfulness guard appends a citation line whenever the draft answer omits one.

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RAG Retrieval and MMR Ranking — Part 2

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GitHub Workflows and Pull Requests — Part 1

Chapter overview: Pull request review starts with listing open PRs by state, reading the changed files and comments, and checking commit history on the branch before approving or requesting changes.

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Key practice: Issue triage labels incoming reports by area and severity, and creating an issue always requires explicit human confirmation because it mutates shared project state visible to the whole team.

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Review checklist: Code search across repositories locates call sites quickly, and reading a file at a pinned ref keeps review comments anchored to exact content.

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Google Workspace Scheduling and Email — Part 1

Chapter overview: Calendar scheduling begins with a free busy availability check, presents the proposed event for human confirmation, and only then creates the meeting on the selected calendar.

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Key practice: Gmail drafting composes a message from meeting context, shows the full payload in a second confirmation gate, and sends only after approval so no external mail leaves without review.

This section expands on the chapter topic with concrete implementation detail, operational guidance, and worked examples drawn from production deployments. Readers should pay attention to configuration defaults, failure modes, and the observability hooks that make the system debuggable under load. Each subsection closes with a checklist that teams can adopt directly in code review. This section expands on the chapter topic with concrete implementation detail, operational guidance, and worked examples drawn from production deployments. Readers should pay attention to configuration defaults, failure modes, and the observability hooks that make the system debuggable under load. Each subsection closes with a checklist that teams can adopt directly in code review. This section expands on the chapter topic with concrete implementation detail, operational guidance, and worked examples drawn from production deployments. Readers should pay attention to configuration defaults, failure modes, and the observability hooks that make the system debuggable under load. Each subsection closes with a checklist that teams can adopt directly in code review.

Review checklist: OAuth tokens are cached locally and refreshed automatically, with a reauthentication URL printed whenever the refresh token expires.

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Google Workspace Scheduling and Email — Part 2

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Google Workspace Scheduling and Email — Part 3

Chapter overview: OAuth tokens are cached locally and refreshed automatically, with a reauthentication URL printed whenever the refresh token expires.

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